AI Finance Assistant: Agentic Financial Intelligence

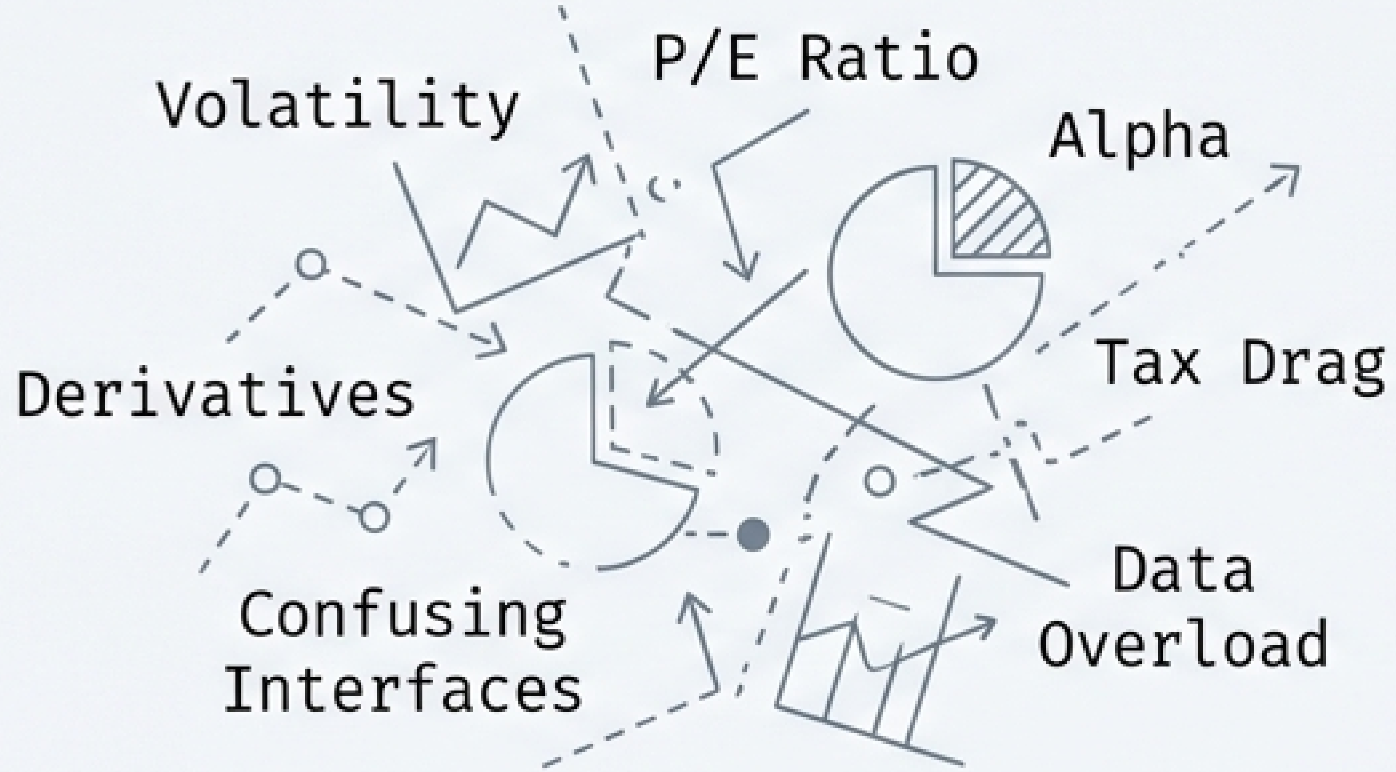
A Production-Grade Multi-Agent System for Financial Literacy & Analysis

Mission: Democratizing financial literacy by combining educational RAG (Retrieval-Augmented Generation) with real-time analytics.

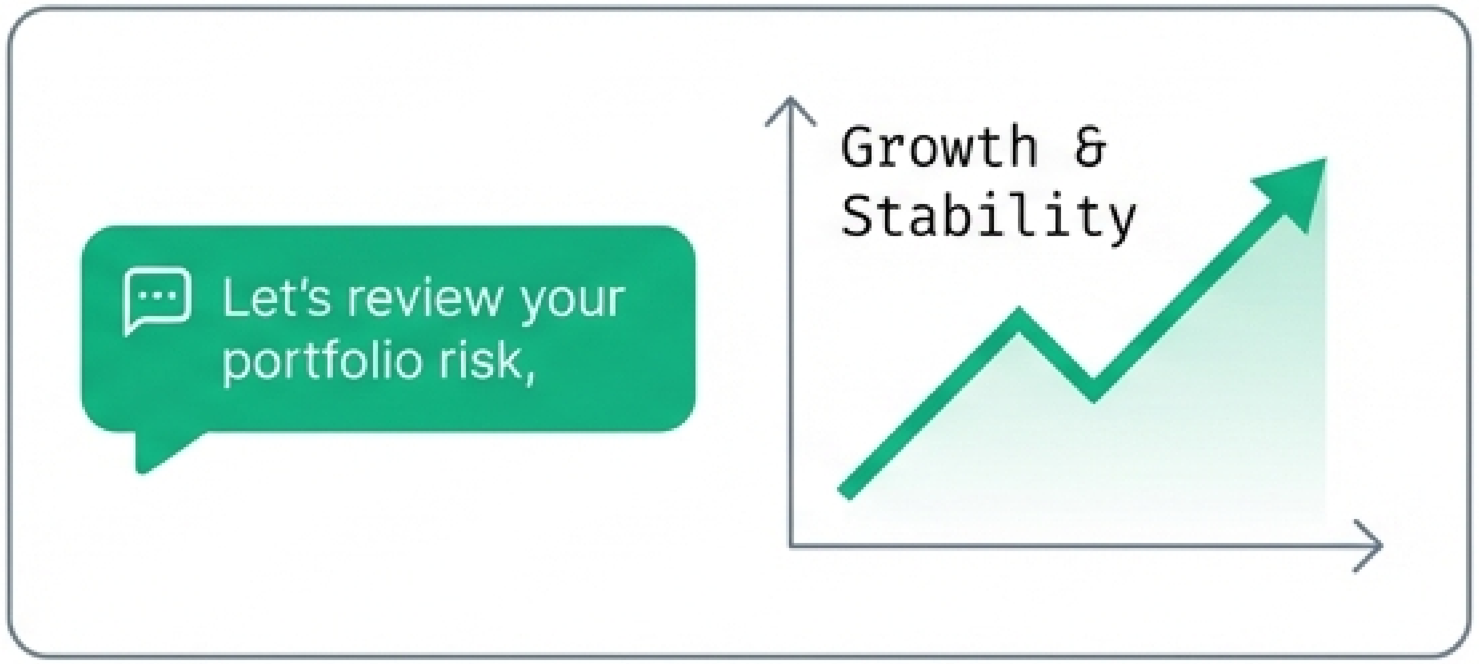
Core Value: Bridging the gap between theoretical knowledge and practical market application through an intelligent, context-aware interface.

The Challenge: Bridging the Financial Literacy Gap

The Barrier



The Solution



Business Opportunity

Democratized Access

Jargon-free explanations for beginners.

Personalized Learning

Adapts to user knowledge levels and risk tolerance.

Real-Time Integration

Learning concepts while seeing live real-world data application.

NotebookLM

The Agentic Workforce

A Coordinated Team of 5 Specialized Agents

Finance QA

The Educator
Uses RAG to answer conceptual questions (e.g., 'What is a Roth IRA?') based on verified articles.

Portfolio

The Analyst
Processes CSV uploads to calculate allocation, risk, and diversification metrics.

Market

The Trader
Fetches real-time prices, volume, and historical trends via yFinance API.

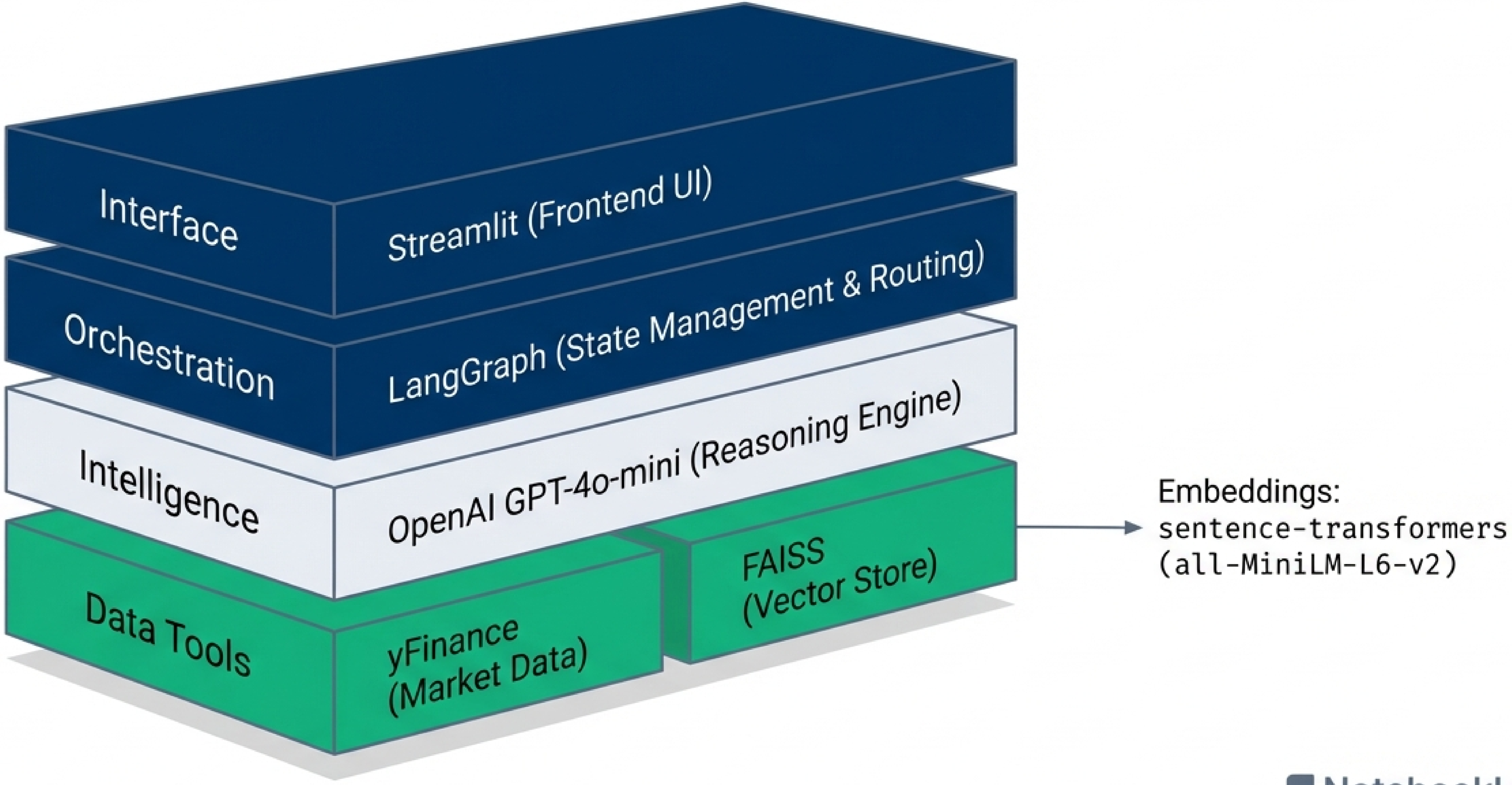
Goal Planning

The Planner
Projects compound interest for retirement or housing targets based on inputs.

News

The Reporter
Aggregates headlines and synthesizes market sentiment from recent news.

Technology Stack & Tooling



Agent Capabilities & Functional Logic

Qualitative & Educational

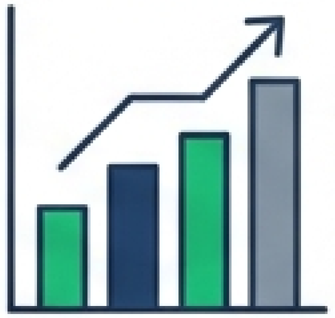
Finance QA Agent

Retrieves context from a vector store (FAISS)

Retrieves context from a vector store (FAISS) using Sentence Transformers embeddings to ground answers in fact.

News Synthesizer

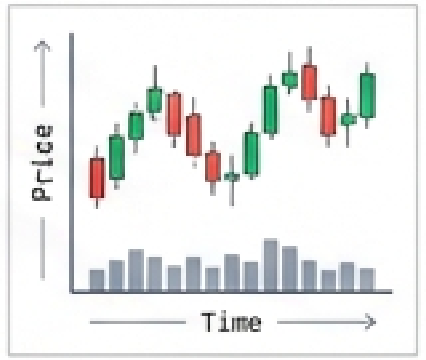
Filters news by topic and generates sentiment summaries to provide market context.



Quantitative & Analytical

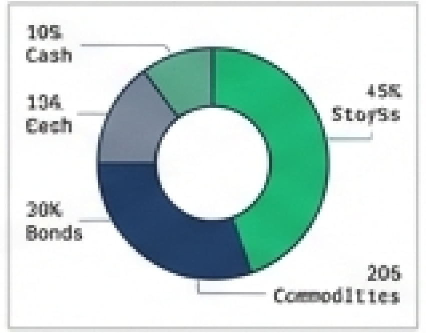
Market Agent

Extracts tickers (e.g., AAPL) to return Price, Volume, and Historical Trends.



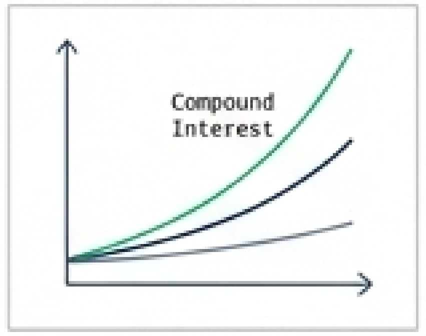
Portfolio Agent

Parses user-uploaded CSVs (ticker, shares, price) to compute Total Value and Asset Allocation %.



Goal Planning

Uses a projection engine to calculate compound interest and required monthly contributions.



User Flow: Context-Aware “Smart Switching”

User Input
How is Apple doing today?

Router
Detects Market Intent

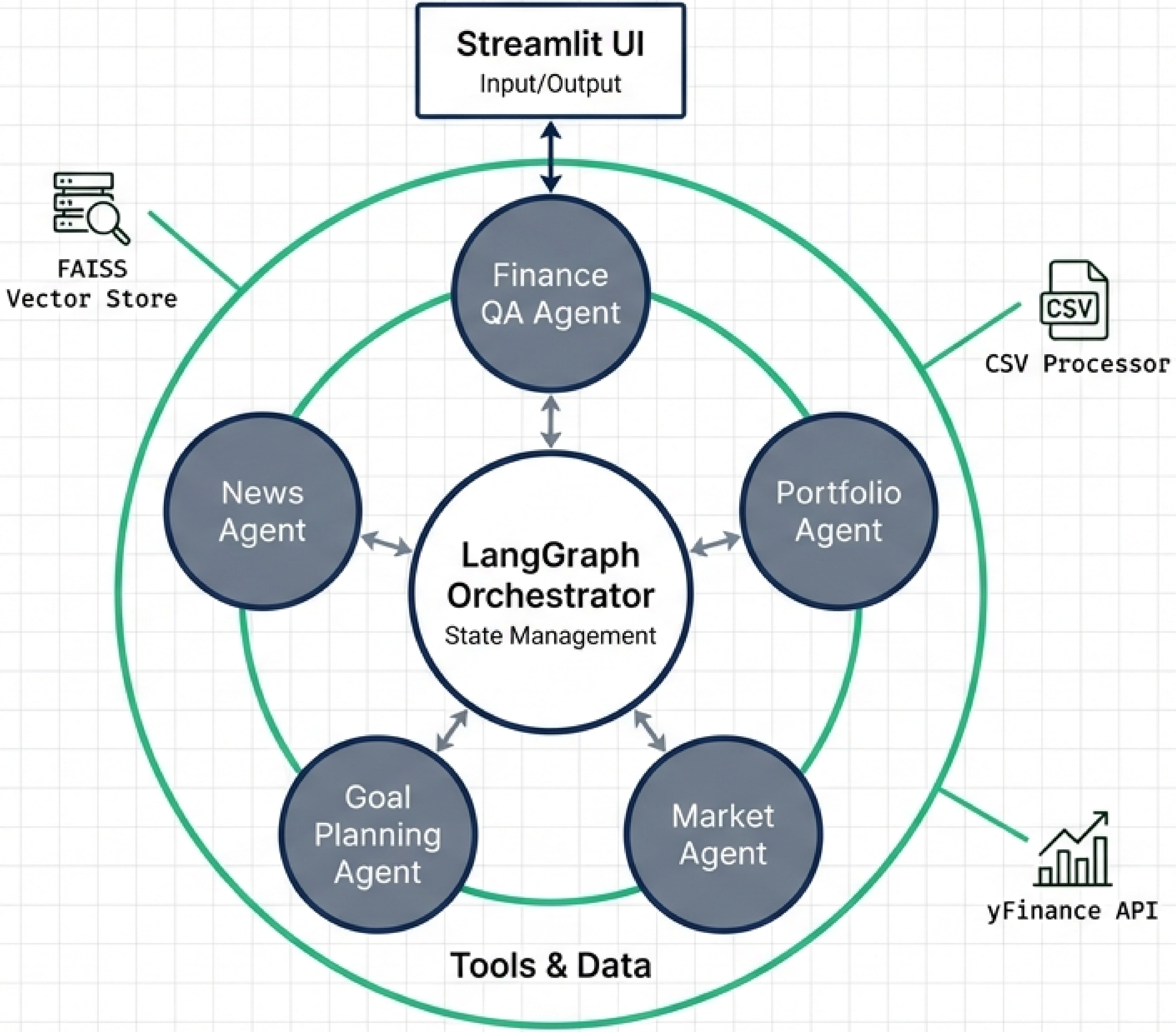
Market Agent
Loads agent to graph

API Call
`yFinance('AAPL')`

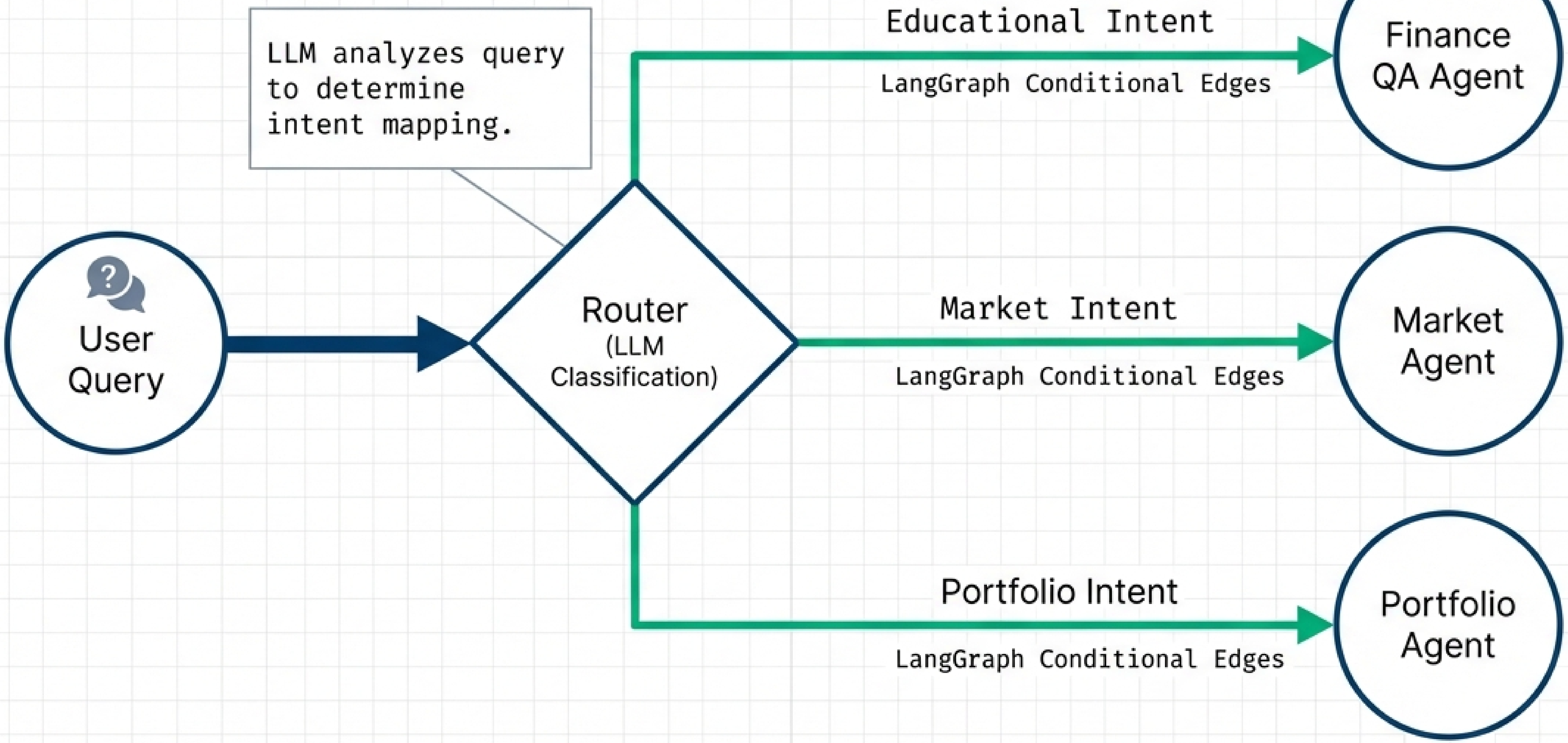
Streamlit UI
Switch tab to ‘Market’ & render chart

Key Innovation: The system bridges the gap between conversation and data visualization by automatically detecting intent and pre-loading the specific UI dashboard.

High-Level System Architecture



Intelligent Routing & Orchestration

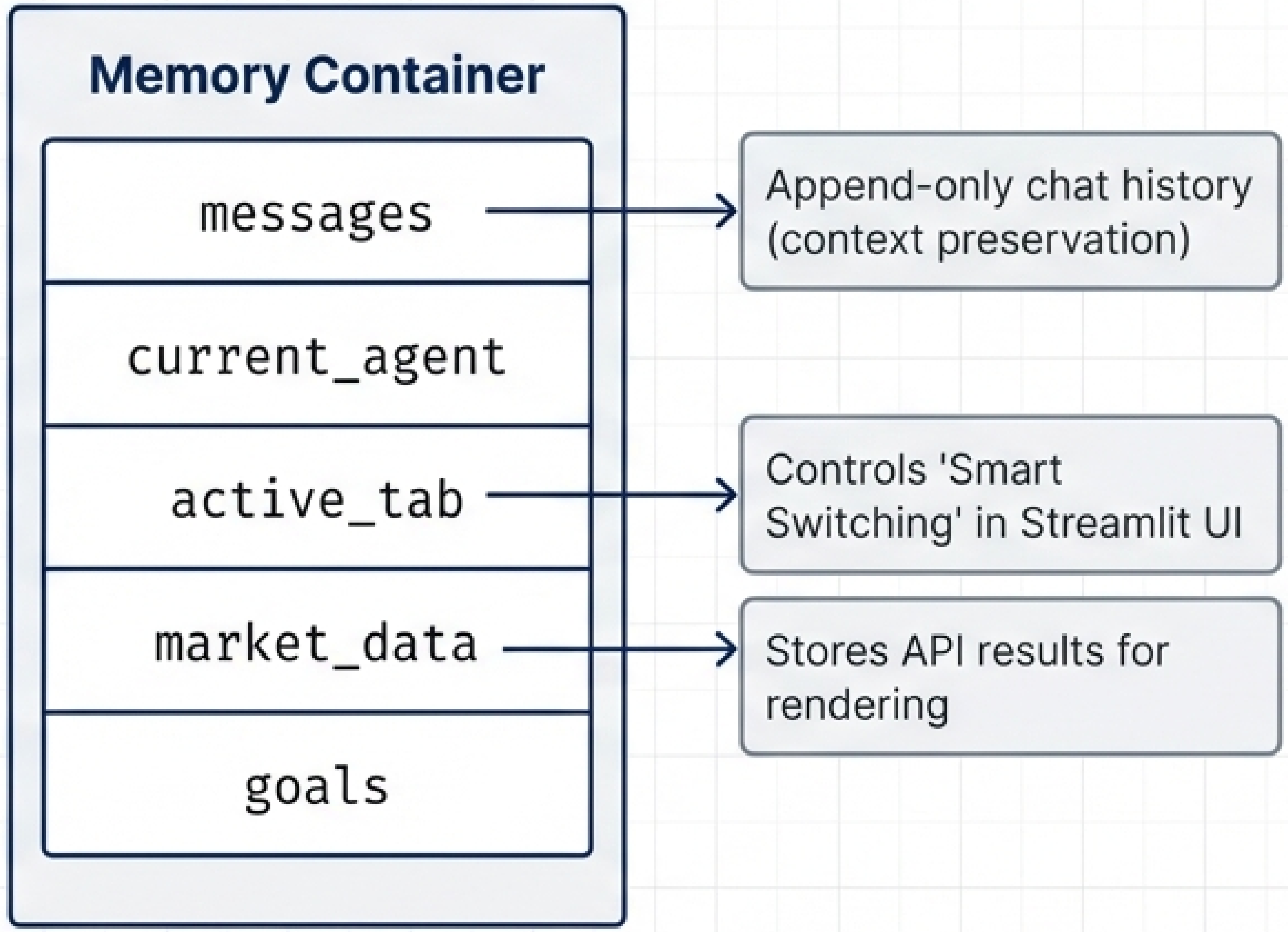


NotebookLM

Deep Dive: The State Object

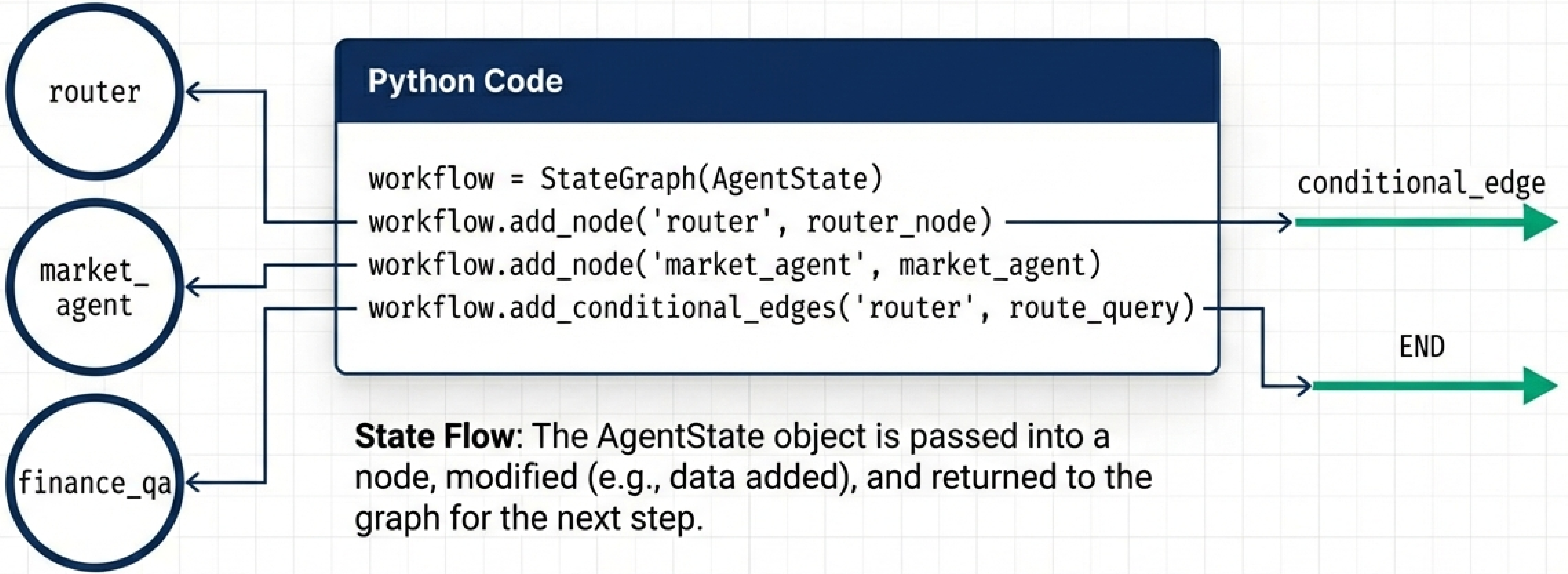
Python Code

```python
class AgentState(TypedDict):
    messages: Annotated[list[BaseMessage],
operator.add]
    current_agent: str
    active_tab: str
    market_data: dict
    goals: list[dict]
```



Deep Dive: Graph Construction

Defining Workflow Logic in Code



State Flow: The AgentState object is passed into a node, modified (e.g., data added), and returned to the graph for the next step.

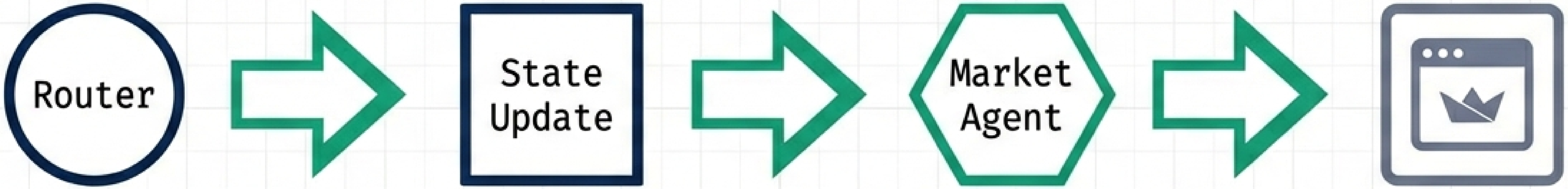
Nodes (Functions)

Edges (Transitions)

Execution Trace: Market Analysis Example

User asks: “How is Apple doing?”

1 2 3 4



1 — Router

Classifies intent as “market_intent”

2 — Fira Code

Sets active_tab = “market”

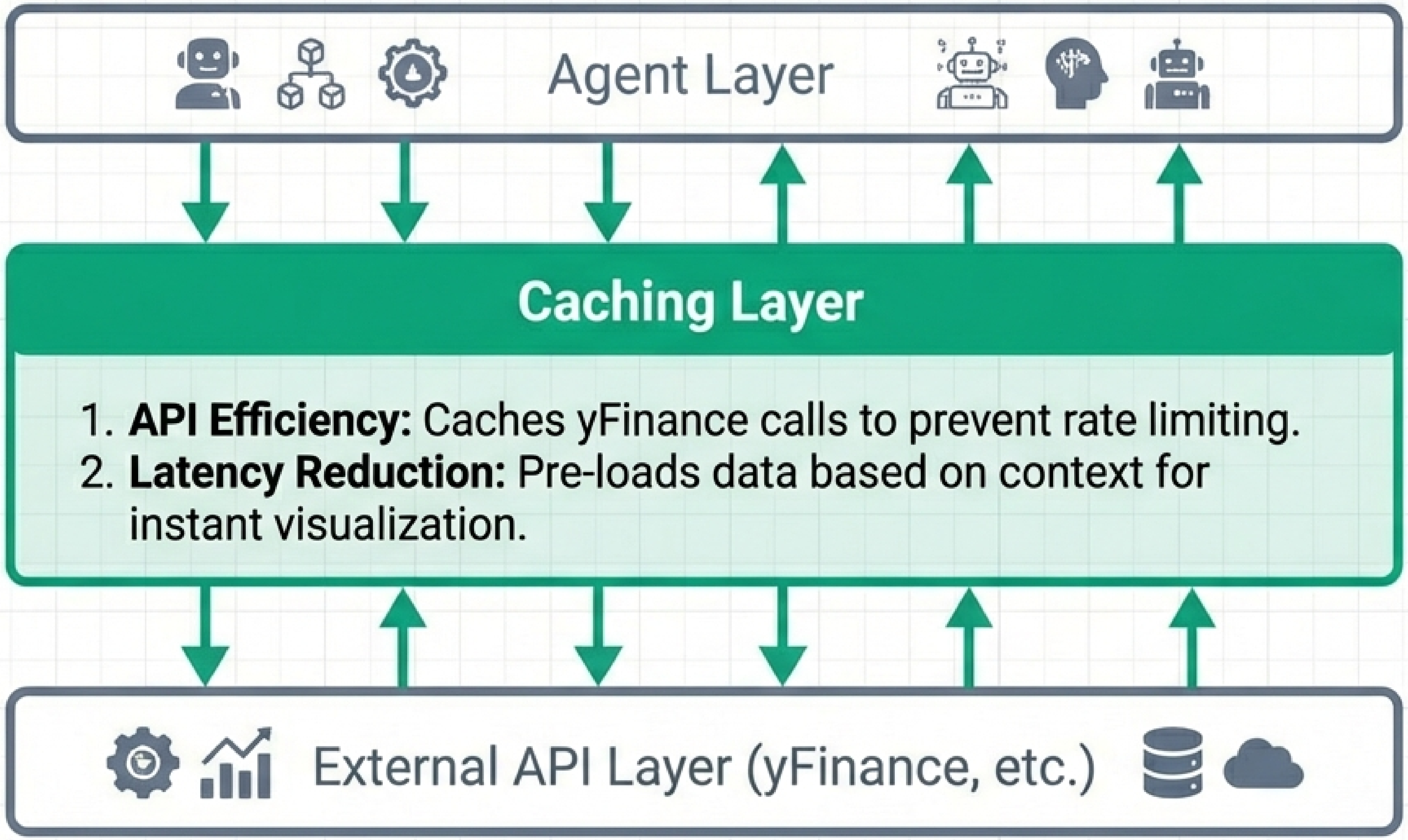
3 — Market Agent

Extracts “AAPL” –> Calls yFinance -> Stores result in state[“market_data”]

4 — UI Render

Streamlit reads active_tab, switches view, renders price chart

Performance & Optimization



Model Optimization

Utilizes GPT-4o-mini for routing and summarization tasks to maintain high speed and low cost without sacrificing reasoning capabilities.

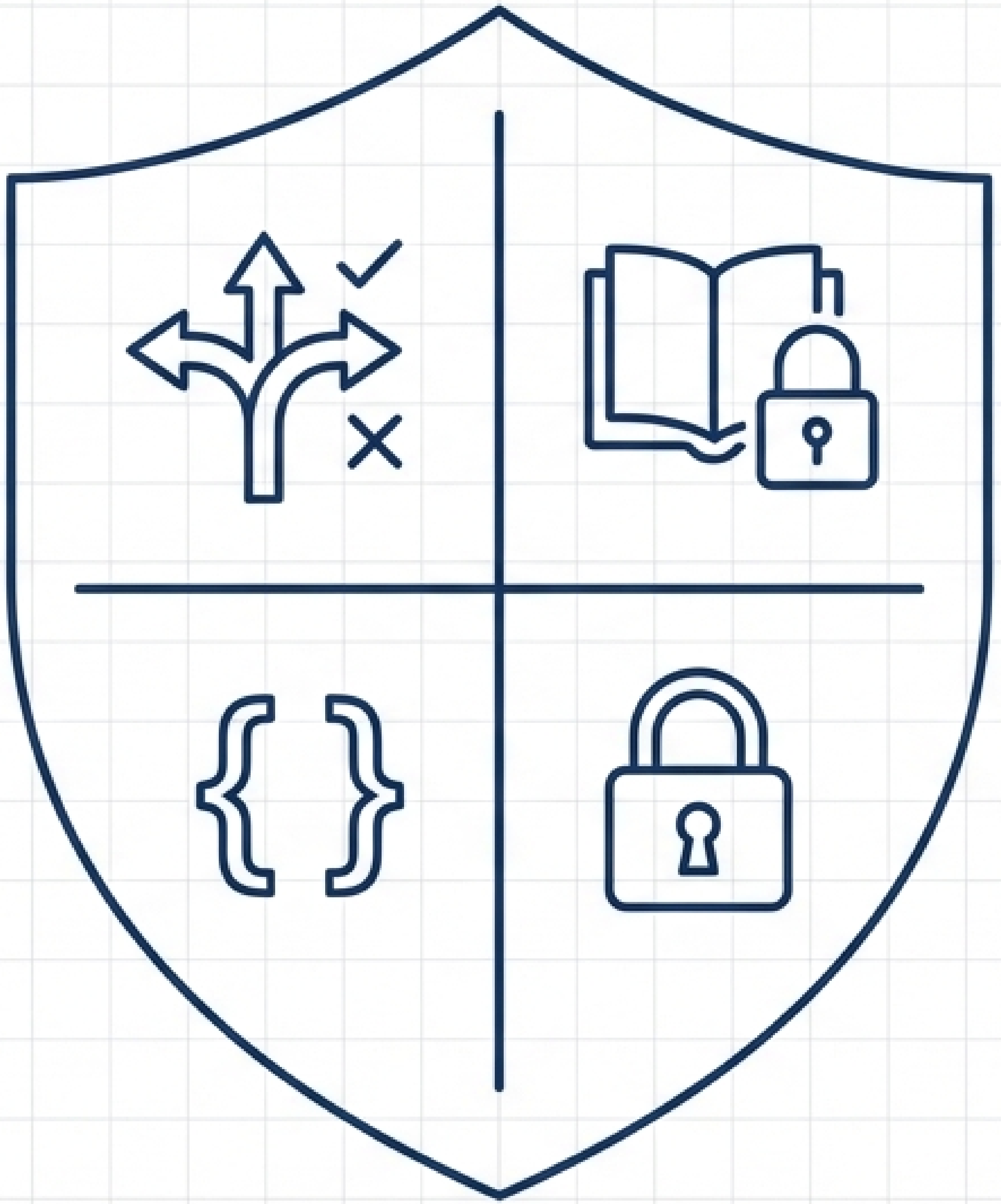
Guardrails & Security

Topic Routing

Prevents Goal Agent from answering Market questions.

Type Safety

Pydantic & TypedDict ensure data integrity.



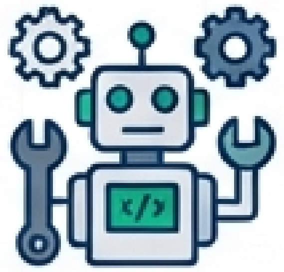
RAG Grounding

QA answers restricted to Knowledge Base context.

Security

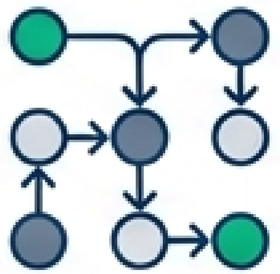
API keys managed via .env, never hardcoded.

Project Impact & Summary



Agentic Power

Moved beyond simple chat to **tool-using agents** (APIs, UI Control).



Architectural Win

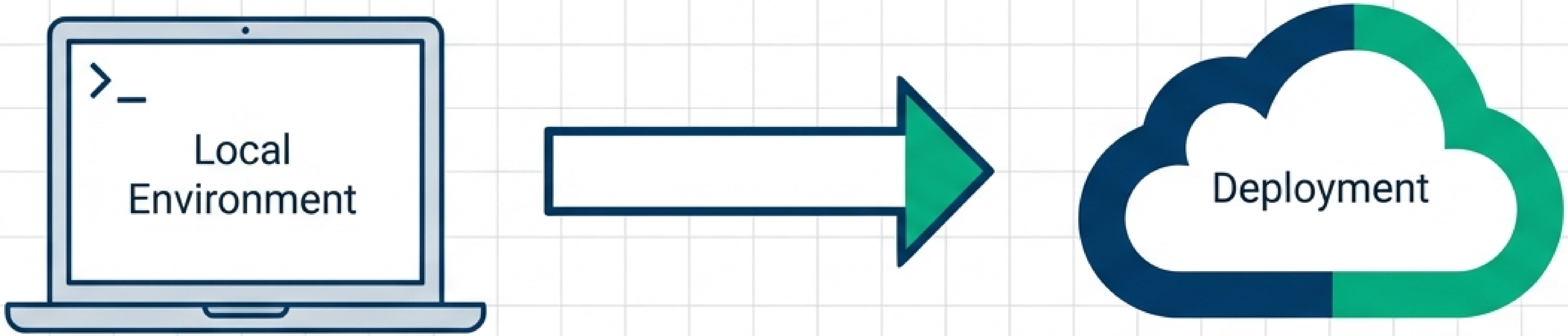
Implemented **LangGraph** state management for complex multi-turn flows.

User Value

Seamless flow between **learning (QA)** and **acting (Market/Portfolio)**.

View the code: github.com/pushkarbh/ai_finance_assistant

Future Roadmap: Cloud Deployment



Containerization

- Full Docker & Docker Compose support.
- Consistent environments (macOS/Linux/Windows).

Next Steps & Expansion

- **Next Steps**: Deploy containerized app to cloud provider (AWS/GCP/HuggingFace Spaces).
- **Scaling**: Implement multi-user authentication and session management.
- **Expansion**: Support for Crypto, Options, and deeper risk modeling.